



Design Your Own Li-Fi DONGLE And SPEAKER

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Light Fidelity (Li-Fi) is a new paradigm in wireless communication. In this article, we describe the design of a Li-Fi dongle (transmitter) and speaker (receiver) for audio and music applications. The proposed system is a replacement for wired speakers that plays audio signals received through light. The dongle can be designed for less than ₹ 350.

This Li-Fi system scores over Bluetooth speakers as follows:

1. The LED light in a room is used not just to light up the room but also to play music, thus giving

benefits of both the worlds. Also, LED bulbs use 85 per cent less energy than incandescent bulbs and last up to 20 times longer, whereas majority of Bluetooth modules are battery powered and generally last a year or two only before they need replacement. Hence the proposed system is energy-efficient over Bluetooth version.

2. You can receive Li-Fi data as long as you are in the range of the light being emitted by the LED light source. Hence, the range depends on the strength of the light being emitted.

3. Connectivity needs to be established between the Bluetooth transmitter and the receiver, whereas with Li-Fi any user can receive data and play music on his speaker by simply keeping the receiver in the range of LED light source.

4. The proposed system can be used with any audio system by just connecting the Li-Fi dongle to its audio output jack, with no pairing and connectivity requests. This overcomes the problem of connecting a Bluetooth

speaker to audio systems that do not have USB port and Bluetooth support, which is otherwise a tougher task.

5. The proposed Li-Fi dongle and speaker costs lower than a Bluetooth dongle and speaker.

The proposed concept is illustrated in Fig. 1. It uses a Li-Fi dongle to transmit audio signals from a source such as mobile phone, and a Li-Fi speaker with solar cells at the receiver end to receive audio signals without wires.

The proposed Li-Fi dongle is connected to the audio jack output available on mobile phones. It has an LED through which audio/music signals are transmitted as light signals. These light signals are captured by the Li-Fi speaker through the solar cell array. The authors' prototype is shown in Fig. 2.

Circuit and working

Fig. 3 shows the circuit of the Li-Fi dongle. It uses three common-emitter (CE) amplifiers connected in parallel to amplify the input audio signal. The Li-Fi speaker is designed by connecting the output of a 3V, 200mA solar cell array to a speaker's audio input. For this,

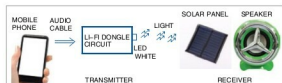


Fig. 1: The proposed Li-Fi based audio system

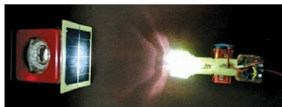


Fig. 2: Author's prototype of Li-Fi transmitter and receiver for audio applications

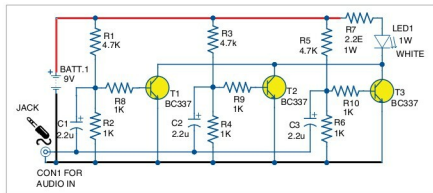


Fig. 3: Circuit of the Li-Fi dongle

PARTS LIST

Semiconductors:

T1-T3 - BC337 npn transistors

LED1 - White LED, 1W

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

R1, R3, R5 - 4.7-kilo-ohm

R2, R4, R6, R8, R10 - 1-kilo-ohm

R7 - 2.2-ohm, 1W

Capacitors:

C1-C3 - 2.2 μ F, 25V electrolytic

Miscellaneous:

CON1 - 2-pin connector

BATT.1 - 9V battery

- Stereo/mono audio jack*

- Li-Fi speaker*

- 3V, 200mA solar panel*

- Audio source (like MP3 player or mobile phone)*

*Not shown in Fig. 3